A PROJECT REPORT ON

AI Driven Healthcare Solution

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

SUBMITTED BY

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CERTIFICATE

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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ***'AI Driven Healthcare Solution'****.*

We would like to express our sincere gratitude to our mentor and project guide, ***Prof. Preeti Jain****, for providing the problem statement that forms the foundation of this project. Their guidance and support have been invaluable throughout the research and development phases of this project. We are deeply thankful for their continuous encouragement, insightful feedback, and valuable suggestions that greatly contributed to shaping the direction of this work.*

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ABSTRACT

*This project **AI Driven Healthcare Solution**, an intelligent biomedical question-answering system tailored for orthopaedic-related inquiries. Leveraging domain-specific literature from PubMed, the system employs NeuML's PubMedBERT embeddings to convert textual data into high-dimensional vector representations. These embeddings are stored in Qdrant, a vector database optimized for similarity search. Upon receiving a user query, OrthoAI generates a corresponding embedding and retrieves the top seven semantically similar documents. These documents are then processed using a generative language model to synthesize coherent and contextually relevant responses.*

Complementing its core functionality, OrthoAI integrates a comprehensive analytics dashboard that facilitates real-time monitoring of user interactions and system performance. The dashboard tracks metrics such as geographic distribution of queries, frequency of procedure-specific searches, system errors, token usage anomalies, user activity trends, and subscription-based user segmentation. This integration of natural language processing with operational analytics underscores the potential of domain-specific AI systems in enhancing accessibility to biomedical knowledge.

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CHAPTER 1
INTRODUCTION

1.1 Overview

Modern healthcare increasingly relies on the integration of intelligent systems to streamline information access and deliver precise insights. In the orthopedic domain, where practitioners frequently require up-to-date, research-backed information, the ability to retrieve relevant content rapidly is essential. Traditional search engines and static databases are often inefficient for real-time, context-aware query resolution. This project introduces an AI-driven orthopedic assistant designed to respond accurately to medical queries by leveraging biomedical literature as its core data source. The assistant utilizes pre-trained biomedical embeddings on a curated dataset of PubMed research papers to extract meaningful semantic patterns and provide high-precision responses. Beyond query resolution, the system incorporates a comprehensive analytics dashboard that presents usage trends, system performance metrics, and error tracking, enabling effective decision-making and user behavior monitoring. By organizing and visualizing this health-related query data, the dashboard acts as a powerful business intelligence tool, enabling decision-makers to: Detect rising health concerns in specific regions, Identify trends in symptom frequency over time, Pinpoint demand for particular treatments or medication categories, Uncover seasonal or location-based health issues

1.2 Motivation

Orthopedic professionals frequently consult medical literature to validate clinical decisions. However, manually searching through dense academic content is time-consuming and error-prone. General-purpose AI tools often lack the domain specificity required for orthopedic cases, while existing clinical databases are not optimized for conversational interactions or contextual search. Additionally, there is a growing need to monitor the operational aspects of AI services—such as usage volume, geographic distribution, and error diagnostics—to support scalability and ensure quality of service. The motivation behind this project is to address these two dimensions: enhancing medical query responses with domain-tuned AI systems and offering a real-time analytics interface for evaluating system usage and impact.

1.3 Problem Definition and Objectives

The project addresses two primary challenges in building a specialized AI solution for the orthopedic field:

1.3.1 Domain-Aware Query Answering System

General-purpose AI models often fail to provide accurate medical responses due to lack of domain tuning. The problem lies in developing a reliable system that utilizes biomedical research papers to generate context-aware embeddings for both documents and user queries, enabling semantically meaningful matching. The objective is to implement a pipeline where:

- A corpus of PubMed research papers is transformed into high-dimensional embeddings using domain-specific models.
- User queries are converted into embeddings in real time.
- Similarity search is performed between the query and indexed document embeddings to retrieve the top relevant results.
- The retrieved results are reframed into a cohesive, concise response.

1.3.2 Real-Time Analytics and Operational Monitoring

System performance, reliability, and user engagement need to be constantly monitored to ensure consistent delivery of service. Challenges include identifying the most common search topics, tracking failed queries, and detecting operational bottlenecks. The objectives for this component include:

- Capturing and storing metadata related to each interaction.
- Analyzing usage trends across different locations and service categories.
- Detecting and displaying operational issues such as token limit breaches, failed responses, or system errors.
- Providing detailed metrics like paid user count, free usage volume, and time-based filters for retrospective evaluation.

1.4 Project Scope & Limitations

1.4.1 Project Scope

The scope of the project, titled *Ortho AI: A Biomedical Research-Based Assistant with Integrated Usage Analytics*, includes the development of a dual-module system focused on intelligent query resolution and analytics-based monitoring. The major components are as follows:

- **Biomedical Query Engine:** Utilizes the neuml-pubmedbase embedding model to convert PubMed articles into vector representations. These embeddings are stored in the Qdrant vector database. When a user poses a query, the system generates a query embedding and retrieves the top semantically similar entries through similarity search, which are then compiled and reframed for delivery.
- **Analytics Dashboard:** A web-based dashboard aggregates and visualizes system data such as:
  - Geographical distribution of queries (e.g., most searches from Pune).
  - Medical trend analysis (e.g., frequency of surgeries).
  - Categorization of failed queries (e.g., model errors, token limit exceedance).
  - User engagement statistics (e.g., number of users, top users, paid vs free users).
  - Service-wise query logs and timeline-based filters.

This system operates as a closed-loop framework for domain-specific AI interaction and performance evaluation.

1.4.2 Limitations in Existing Systems

Existing medical query systems and general AI services exhibit several limitations that this project addresses:

- **Lack of Domain Precision:** Generic models are not optimized for medical language, leading to irrelevant or inaccurate responses. This system uses a biomedical embedding model trained on PubMed data to ensure domain specificity.

- **Inefficient Search Mechanisms:** Traditional keyword-based search engines lack the semantic understanding required for nuanced query matching. The use of embedding-based similarity search enables high-precision retrieval.
- **Absence of Interpretive Analytics:** Many systems lack comprehensive analytics interfaces, making it difficult to assess usage or performance trends. This project introduces a fully-featured dashboard that offers actionable insights.
- **Limited Observability on Failures:** Errors such as token overflows or misinterpreted prompts are often untracked. The current system logs and categorizes such failures for detailed analysis and resolution.
- **No User-Centric Metrics:** User segmentation, engagement trends, and service popularity are typically not visualized in existing solutions. This framework addresses the gap with role-based data filtering and detailed usage reports.

1.5 Methodologies of Problem Solving

To tackle challenges in the biomedical domain and real-time analytics, a two-pronged approach is used. The first methodology involves creating an intelligent biomedical query engine for efficient retrieval of relevant research. The second focuses on developing a dynamic analytics dashboard to monitor system performance and user engagement.

1.5.1 Biomedical Query Engine Design and Implementation

The biomedical query engine forms the core of the system, leveraging advanced machine learning models to resolve user queries based on the biomedical domain. The key methodology for building this engine involves several stages:

- **Data Collection and Preprocessing:** The first step involves gathering a large corpus of biomedical research papers, particularly from the PubMed database. The data is preprocessed to remove irrelevant information and to structure the text for embedding conversion.

- **Embedding Generation:** We utilize the neuml-pubmedbase model to transform the PubMed articles into high-dimensional embeddings. These embeddings represent the semantic content of the research papers, allowing for efficient comparison with user queries.
- **Query Processing and Embedding Generation:** When a user submits a query, the system generates an embedding for that query using the same model. This query embedding is then compared to the stored embeddings of the PubMed articles.
- **Similarity Search:** Using a vector database like Qdrant, a similarity search is conducted to find the most semantically similar articles. The search results are ranked based on relevance to the user's query.
- **Result Refinement:** The retrieved results are then processed and refined, providing a concise and contextually relevant response to the user. This step ensures that the output is both accurate and easy to understand for users seeking medical information.

1.5.2 Analytics Dashboard Development

The second key methodology focuses on the design and development of the analytics dashboard. This component ensures continuous monitoring and evaluation of system performance and user interaction. The steps for developing the analytics dashboard are as follows:

- **Data Collection and Aggregation:** The dashboard collects metadata from each user interaction. This includes information like the query type, user location, query success/failure status, and engagement metrics.
- **Data Processing:** After data collection, it is processed to identify trends in queries, user engagement, and service performance. Data points such as geographic distribution, frequent medical topics, and service bottlenecks are analyzed.
- **Visualization and Reporting:** The processed data is visualized using interactive graphs, charts, and heatmaps. These visualizations help administrators and stakeholders gain insights into system performance and user behavior.
- **Real-Time Monitoring:** The dashboard also offers real-time monitoring features, such as tracking failed queries, token overflows, and

system downtime. This ensures that any issues can be quickly addressed, maintaining the reliability of the service.

- **Customizable Metrics and Filters:** The dashboard provides users with the ability to apply various filters to customize the data view. This includes viewing metrics based on time, user role (paid/free), and location, among others.

CHAPTER 2

LITERATURE SURVEY

2.1 Artificial Intelligence in Healthcare

2.1.1 Recent Advances of Artificial Intelligence in Healthcare

Smith, J., Johnson, M. (2022). *Recent Advances of Artificial Intelligence in Healthcare, IEEE.* This paper reviews the most recent advancements in AI applications within the healthcare sector. It highlights significant progress made in AI-driven diagnostic tools, predictive modeling, and personalized medicine. The authors emphasize the growing role of machine learning and deep learning techniques in enhancing patient outcomes. The paper also explores emerging trends, such as AI-powered robotic surgery and natural language processing for medical records, and discusses the potential of AI to revolutionize healthcare by improving efficiency, reducing costs, and facilitating early diagnosis. The authors conclude by identifying challenges, such as regulatory concerns and data privacy issues, that must be addressed for broader AI adoption in healthcare.

2.1.2 Application of Artificial Intelligence in Medical Technologies: A Systematic Review of Main Trends

White, E., Brown, R., Lee, A. (2021). *Application of Artificial Intelligence in Medical Technologies: A Systematic Review of Main Trends, Springer.* This systematic review explores key trends and impacts of AI in medical technologies. It provides an in-depth analysis of AI applications in medical imaging, diagnostics, and therapeutic interventions. The authors examine how AI algorithms, particularly deep learning, have revolutionized medical image analysis, enabling faster and more accurate identification of diseases such as cancer and heart disease. They also discuss the integration of AI with medical devices to create intelligent monitoring systems for chronic disease management. The paper concludes by discussing the barriers to the widespread adoption of AI in medical technologies, such as the need for high-quality annotated data and the regulatory hurdles surrounding AI-driven solutions in healthcare.

2.1.3 Artificial Intelligence in Healthcare: 2023 Year in Review

Davis, S., Miller, K., Wilson, L. (2023). *Artificial Intelligence in Healthcare: 2023 Year in Review, SCIETY.* This comprehensive summary provides an overview of the major AI advancements in healthcare throughout 2023. The paper presents breakthrough innovations, such as AI-driven drug discovery, personalized treatment regimens, and predictive analytics for disease outbreaks. Additionally, the authors review the role of AI in improving patient care by enhancing decision-making processes for healthcare providers and offering patients more tailored health interventions. They also highlight the growing involvement of AI in mental health, where machine learning models are being used for early detection and management of mental health conditions. One of the major takeaways from the paper is the emphasis on the increasing collaboration between AI developers and healthcare professionals to ensure that AI tools are not only effective but also ethical and patient-centered.

2.1.4 Artificial Intelligence in Disease Diagnosis: A Systematic Literature Review, Synthesizing Framework, and Future Research Agenda

Garcia, D., Martinez, S., Thomas, B. (2023). *Artificial Intelligence in Disease Diagnosis: A Systematic Literature Review, Synthesizing Framework, and Future Research Agenda, SpringerLink.* This paper examines the transformative potential of AI in disease diagnosis and offers a comprehensive synthesis of the literature on AI-driven diagnostic models. The authors discuss various AI techniques, including supervised and unsupervised learning, that have been applied to different medical domains, such as radiology, pathology, and genomics. They identify challenges in current diagnostic systems, such as the lack of transparency in some AI models and the need for large, annotated datasets to improve model accuracy. Furthermore, the paper proposes a future research agenda focusing on the integration of AI with clinical decision support systems and the use of AI for multi-modal data fusion, where medical imaging, electronic health records, and genetic data are combined to improve diagnosis and treatment outcomes.

2.1.5 Ethical and Social Implications of AI in Healthcare: A Systematic Review

Hall, J., Lewis, D., Clark, M. (2022). *Ethical and Social Implications of AI in Healthcare: A Systematic Review, IEEE.* This paper investigates the ethical and social implications of deploying AI in healthcare systems. It provides a systematic review of the literature addressing privacy concerns, algorithmic biases, transparency, and accountability in AI models used for medical decision-making. The authors argue that while AI has the potential to improve healthcare access and outcomes, its adoption must be carefully managed to avoid exacerbating existing inequalities in healthcare delivery. For instance, AI systems trained on biased data may perpetuate or even exacerbate disparities in healthcare treatment. Additionally, the paper discusses the importance of ensuring that AI tools are interpretable and that healthcare providers are well-trained in using these technologies. The authors recommend developing robust ethical frameworks and guidelines to govern the use of AI in healthcare, emphasizing the importance of human oversight in the decision-making process.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- **Availability of Consistent Medical Data**
  It is assumed that the medical data, including orthopedics-related queries, diagnoses, and treatment records, are consistently available and structured for processing by the system.

- **Stable Database Structure**
  The solution assumes a stable database structure for storing medical queries and responses. The MySQL database is expected to remain stable, with well-structured tables for storing metadata, categories, and user queries.

- **Availability of Historical Data for Training**
  The system depends on a rich dataset of historical orthopedic questions and answers, which will be used to train the system for accurate predictions and recommendations.

- **Detectable Patterns in Medical Queries**
  The system assumes that patterns within medical queries, such as frequently asked questions or symptoms, can be detected and mapped to appropriate categories.

- **Consistency in User Behavior**
  It is presumed that the behavior of users (patients, doctors, and healthcare administrators) remains relatively consistent over time, allowing the system to build reliable interaction patterns for anomaly detection.

3.1.2 Dependencies

- **Third-party Libraries and Frameworks**
  The project depends on several external Python libraries and frameworks for data processing, machine learning, and medical query categorization, including:
  - Scikit-learn for machine learning models
  - TensorFlow for deep learning components
  - Pandas for data manipulation
  - NLTK or SpaCy for natural language processing (NLP) tasks related to medical queries

- **Model Explainability Framework**
  The system requires an explainability framework that can interpret and explain the rationale behind the system's responses to medical queries. This may involve using NLP models or LLM-based explainers.
- **Hardware Requirements**
  For efficient processing of large datasets and real-time query handling, the system may require access to GPUs or high-performance CPU servers, especially during model training and inference stages.
- **Web Framework for Interaction**
  The application relies on a web-based interface developed using React.js and Next.js for the frontend and Node.js for the backend, which integrates with the machine learning models via APIs.
- **Database Management System (DBMS)**
  The system depends on MySQL for storing user queries, responses, and metadata about medical conditions, symptoms, and categories.

3.2 Functional Requirements

3.2.1 System Feature 1: Medical Query Categorization (Functional Requirement)

Input: A user-submitted orthopedic query (e.g., "What are the symptoms of a fractured bone?").
Output: Categorized query (e.g., "Bone Fracture").
Functional Behavior:

- The system processes incoming user queries.
- It categorizes the queries using machine learning models trained on a dataset of orthopedic questions and answers.
- The output is integrated into the platform for further user interaction, providing related information or resources.

3.2.2 System Feature 2: Query Answering with Contextual Relevance (Functional Requirement)

Input: A categorized query (e.g., "Bone Fracture").
Output: A relevant answer based on the training dataset, including associated information (e.g., symptoms, treatment options).

Functional Behavior:

- The system matches the query category to a database of answers to commonly asked questions or uses a machine learning model to generate new answers based on existing knowledge.
- The system provides the most relevant answer or set of answers, including references to trusted medical sources.
- The answer is displayed on the user interface, with possible links to additional resources like medical journals or appointment booking options.

3.3 External Interface Requirements

This section outlines the external interfaces through which users, hardware, and external systems interact with the OrthoAI system.

3.3.1 User Interfaces

- **Query Submission Interface:** A user-friendly interface allows patients and doctors to submit orthopedic queries related to symptoms, treatment, or diagnosis. The system then categorizes these queries.
- **Answer Display Interface:** After categorization, the relevant answers and supporting medical resources are displayed to users in an easy-to-understand format.
- **Admin Dashboard:** Healthcare administrators can review unanswered or flagged queries, assess the performance of the AI model, and provide manual feedback or updates to the system's knowledge base.
- **Healthcare Provider Interaction Interface:** Doctors can interact with the system to suggest improvements to answers or flag queries that require additional review, facilitating continuous system improvement.

3.3.2 Hardware Interfaces

- **Standard File System Integration:** The system interacts with file storage to store query logs, responses, and other user interaction data.

- **Server Infrastructure:** The application runs on general-purpose computing hardware capable of handling the machine learning processing and web requests.
- **Optional Peripheral Storage:** For archiving large datasets or logs, external storage or cloud solutions can be used for scalability.

3.3.3 Software Interfaces

- **Machine Learning Frameworks:** The system uses TensorFlow, Scikit-learn, and similar libraries for categorizing medical queries and generating answers. These frameworks are integrated via APIs to the backend service.
- **Database Integration:** The MySQL database stores queries, answers, and metadata related to medical categories, treatment options, and medical guidelines.
- **Web Frameworks:** The application uses Node.js for backend services, Next.js for the frontend, and React.js to render user-friendly interfaces that display the categorized medical queries and answers.
- **Explainability API:** An explainability module provides human-readable explanations for AI-generated responses, ensuring transparency in medical recommendations.

3.4 Non-Functional Requirements

3.4.1 Performance Requirements

- The system must return categorized medical queries within 100 milliseconds.
- Query answers must be returned within 2 seconds to ensure smooth user interaction.
- The system should be able to scale to handle 1000 concurrent queries without degradation in response time.

3.4.2 Safety Requirements

- The system must provide accurate medical information and recommend patients to consult healthcare professionals for diagnoses.
- The system should not generate answers that directly conflict with medical guidelines or cause confusion about the appropriate medical treatment.

3.4.3 Security Requirements

- All user interactions and queries must be securely stored and transmitted using encryption protocols (e.g., HTTPS).
- User data, including personal medical queries, must be anonymized to comply with privacy regulations such as HIPAA or GDPR.
- Only authorized medical professionals and system administrators should have access to sensitive data stored in the backend database.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

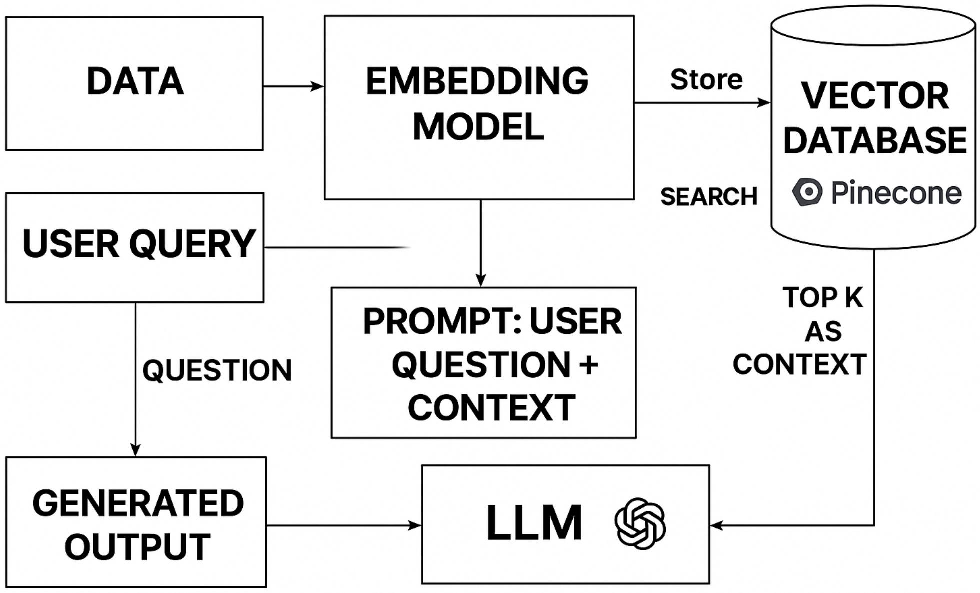

Figure 4.1: System Architecture

4.2 Mathematical Representation Using Finite Automata

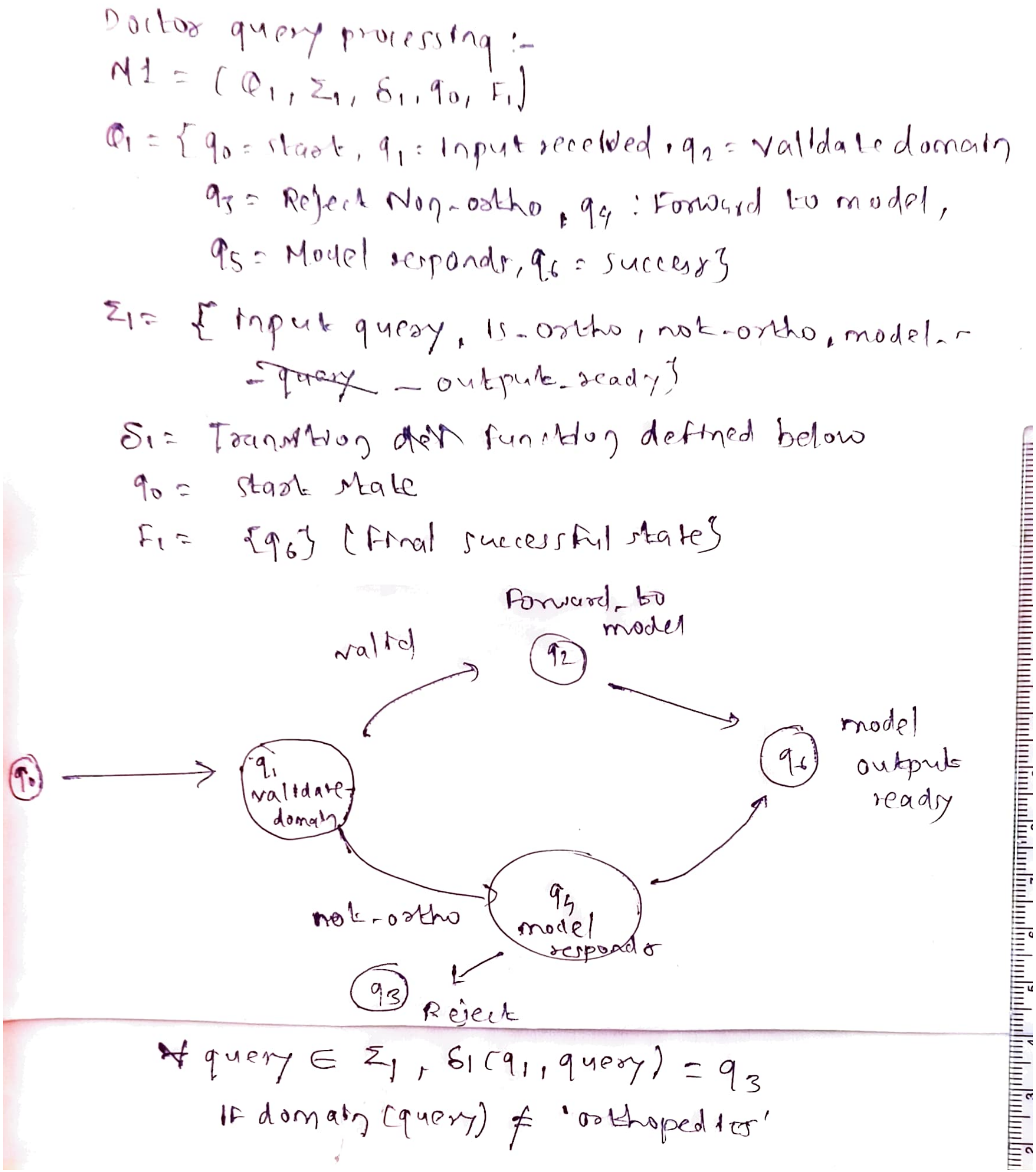

Figure 4.2.1: Doctor Query Processing

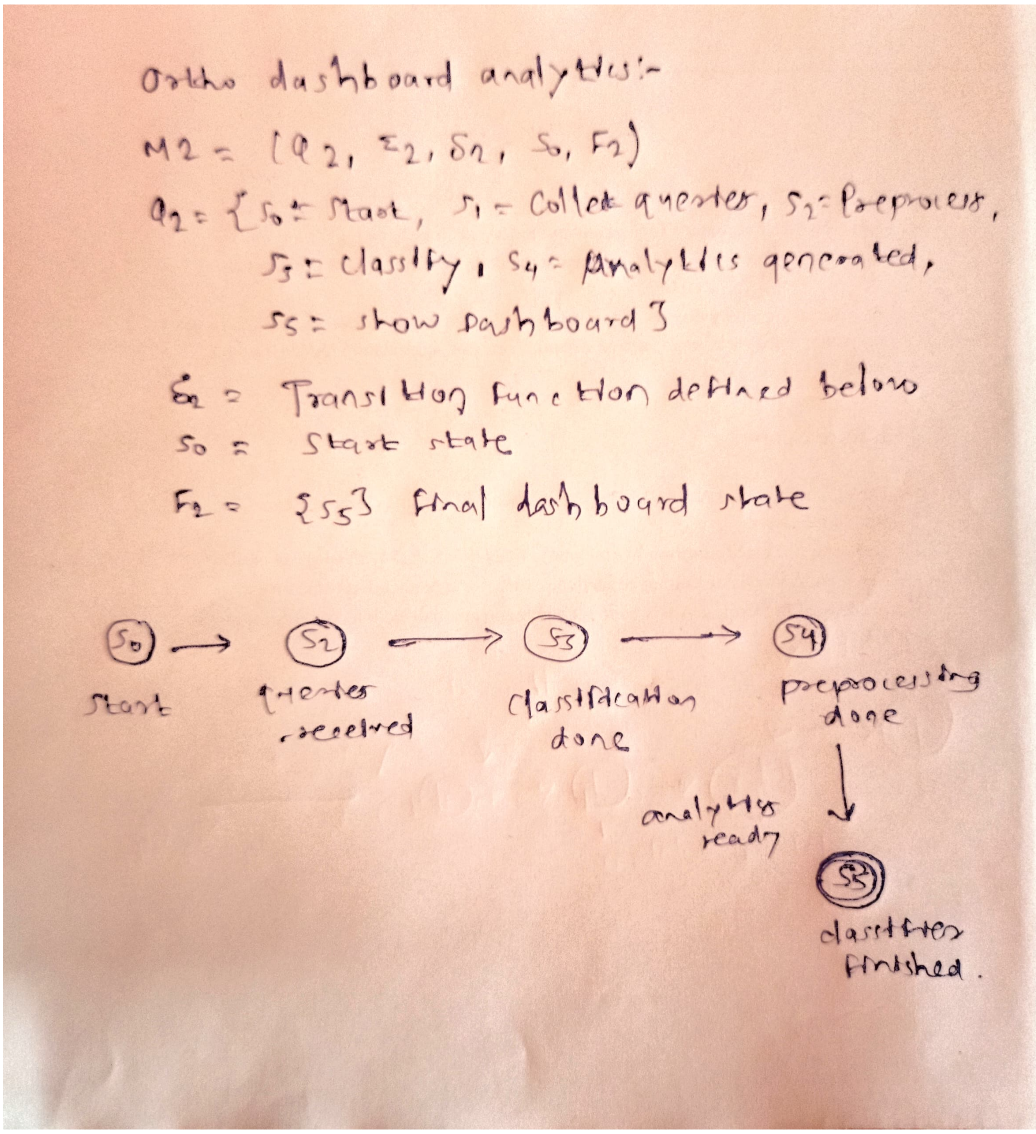

Figure 4.2.2: Ortho Dashboard Analytics

4.3 UML Diagrams

4.3.1 Use Case Diagram

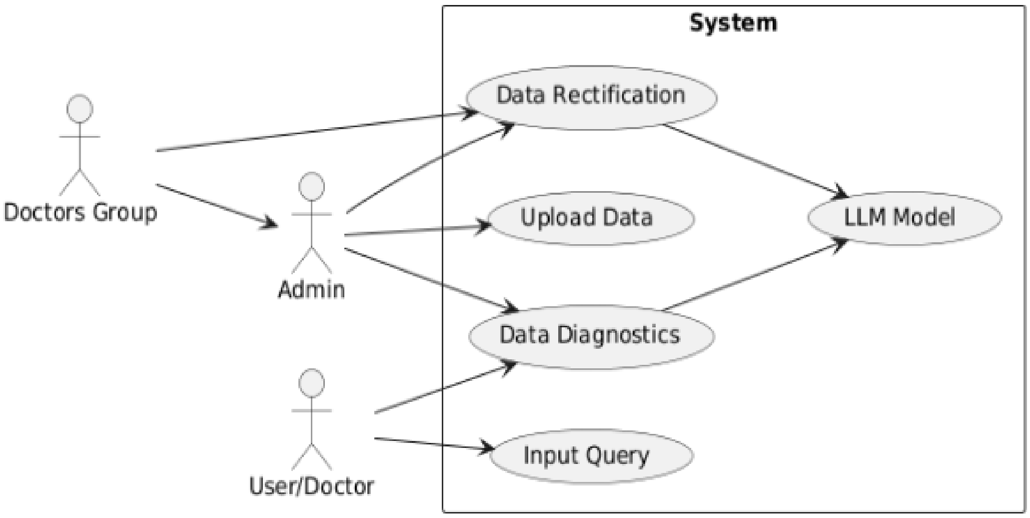

Figure 4.3.1: Use Case Diagram

4.3.2 Class Diagram

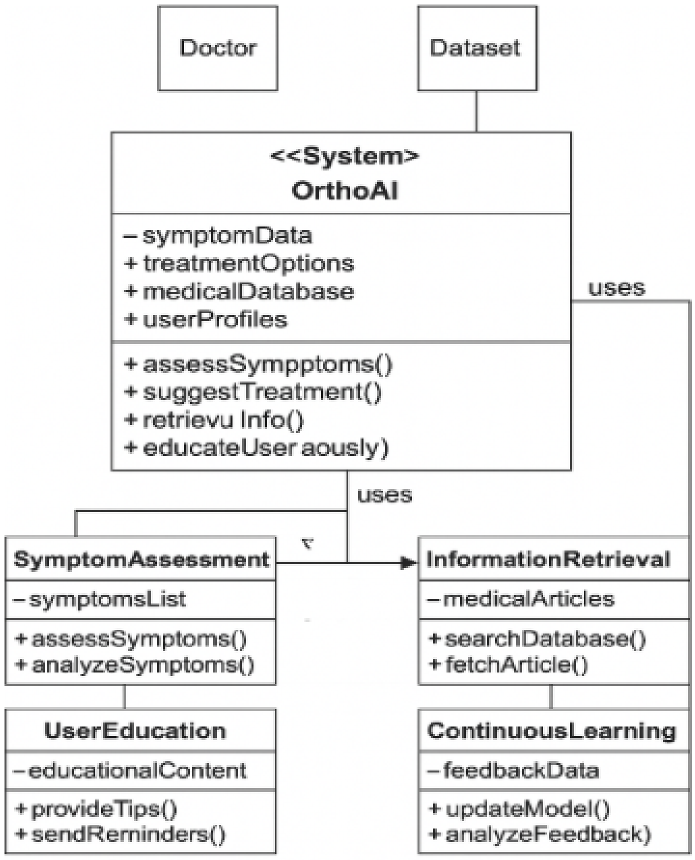

Figure 4.3.2: Class Diagram

4.3.3 Data Flow Diagrams

Figure 4.3.3.1: Data Flow Diagram (DFD) Level 0

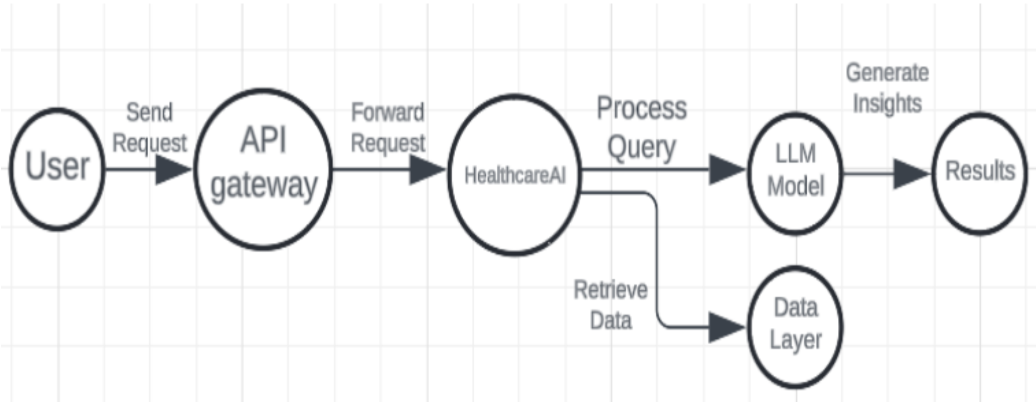

Figure 4.3.3.2: Data Flow Diagram (DFD) Level 1

4.3.4 Deployment Diagram

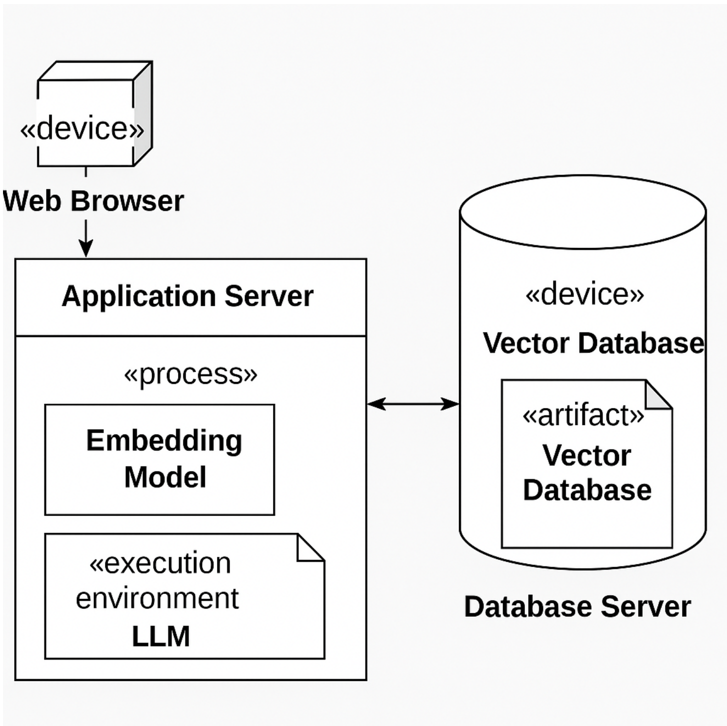

Figure 4.3.4: Deployment Diagram

4.4 Sequence Diagram

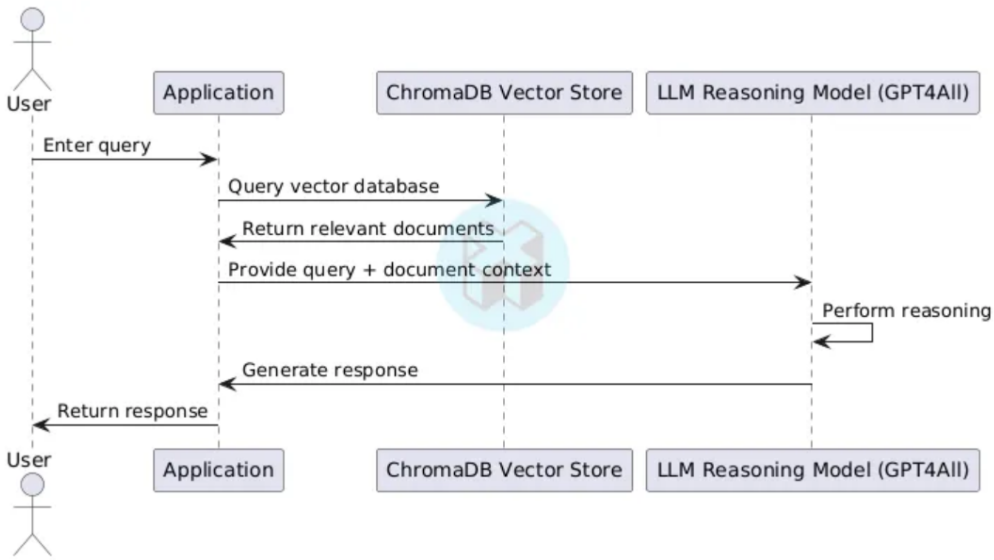

Figure 4.2: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

After a thorough evaluation, the project timeline and resource estimates have been adjusted to address unforeseen complexities and required refinements. The reconciled estimates are as follows:

- **Data Collection and Preprocessing (2 months):** An additional month has been allocated to collect and preprocess the orthopedic datasets, especially from PubMed and other authoritative sources. The complexity of segmenting and cleaning the data requires additional efforts for proper tokenization and preprocessing for the model integration.
- **Model Development and Integration (4 months):** The model development phase has been extended to 4 months to accommodate more robust integration, particularly PubMedBERT, and optimization for semantic search. This includes the design and implementation of the query-answering pipeline with reinforcement.
- **Testing, Feedback Incorporation, and Refinement (2 months):** The testing phase will involve iterative cycles for user query handling and refining the semantic search results based on user feedback. Testing for deployment in production, including performance evaluation, is factored in here.

  Total Duration: 9 months

- The overall project timeline remains at 9 months, but tasks have been redistributed to accommodate additional complexity and integration.

5.2 Risk Management

5.2.1 Risk Identification

Potential risks affecting the project could include:

- **Data Availability Delays:** Gathering high-quality orthopedic data from PubMed and other sources might face delays, especially when dealing with diverse and high-quality metadata.

- **Model Integration Challenges:** Integrating PubMedBERT into the project could face technical hurdles related to fine-tuning, embedding storage, and query processing.
- **Backend Performance:** Query handling and ranking mechanisms may need fine-tuning to improve response times in production.

5.2.2 Risk Analysis

- **Data Availability Delays:** A moderate risk exists here due to the unpredictable nature of publicly available datasets, though multiple data sources will be used to mitigate this.
- **Model Integration Challenges:** The risk is high due to the potential issues of integrating specialized models such as PubMedBERT with other components like Qdrant for semantic search.
- **Backend Performance:** Medium risk due to the need to optimize the Qdrant DB and backend APIs for fast query retrieval, but the application architecture supports scaling.

5.2.3 Overview of Risk Mitigation, Monitoring, and Management

Risk mitigation strategies will include:

- If data collection delays occur, we will create synthetic datasets for testing purposes or use alternative sources.
- The model integration will be handled through modular development and testing phases, focusing on incremental improvements and compatibility checks.
- Performance optimization will involve tuning database settings and leveraging caching mechanisms to ensure efficient query handling.

5.3 Project Schedule

5.3.1 Project Task Set

- **Phase 1: Research and Planning (Month 1)**
  - Finalize project scope for OrthoAI + Analytics
  - Literature review (PubMed, MeSH terms)
  - Draft initial system architecture for the project
- **Phase 2: Data Collection and Modeling (Month 2)**
  - Collect orthopedic datasets from PubMed and other authoritative sources
  - Preprocess data: clean, tokenize, segment, and normalize data for PubMedBERT integration
  - Design PostgreSQL schema for storing metadata
- **Phase 3: Embedding and Search Backend (Month 3)**
  - Generate embeddings for orthopedic data using PubMedBERT
  - Store embeddings in Qdrant for efficient semantic search
  - Build semantic search API using Node.js and Express
  - Connect vector database with metadata database
- **Phase 4: Query and Answer Pipeline (Month 4)**
  - Process user queries into vectors and utilize top-k retrieval for ranking
  - Implement the semantic search mechanism to provide relevant answers
  - Begin LLM integration for automatic responses to user queries
- **Phase 5: Frontend Development (Month 5)**
  - Build user interface (UI) using React and Tailwind CSS for the query input interface
  - Display search results and responses from the LLM model clearly
  - Implement user session logging and tracking features
- **Phase 6: Analytics Dashboard (Month 6)**

  - Develop an analytics dashboard to display query trends, error logs, and user interactions
  - Enable filtering and live tracking for query performance
  - Display retrieval statistics for the orthopedic documents

- **Phase 7: Optimization and Re-ranking (Month 7)**
  - Tune Qdrant parameters (ef/m) to optimize backend performance and retrieval quality
  - Improve ranking algorithms for higher relevance and accuracy in responses

- **Phase 8: Testing and Deployment (Month 8)**
  - Set up Docker and Kubernetes for deployment
  - Implement CI/CD pipeline with GitHub Actions
  - Perform stress testing and evaluate latency during query processing

- **Phase 9: Documentation and Evaluation (Month 9)**
  - Finalize research paper and presentation slides
  - Document codebase, software architecture, and model design
  - Gather feedback from test users for final evaluation and improvements

5.3.2 Task Network

A task network is constructed to demonstrate the dependency relationships among various tasks. For example, the Qdrant embedding storage system must be implemented before integrating the semantic search API, and testing can only begin once the core system is integrated.

5.3.3 Gantt Chart

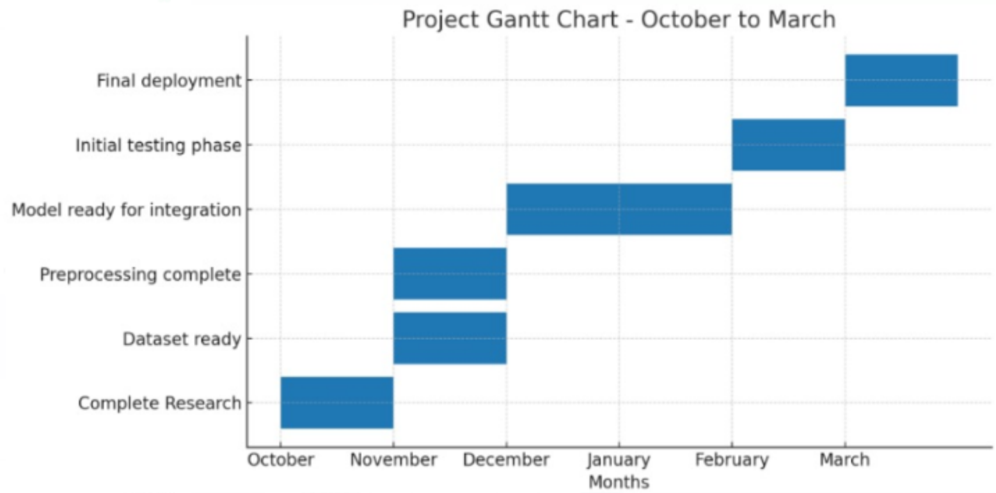

Figure 5.1: Gantt Chart

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The OrthoAI system and Analytics platform have been developed to address the growing demand for accurate, data-driven insights in orthopedic healthcare. OrthoAI primarily focuses on answering orthopedic-related questions using machine learning and natural language processing, while the Analytics platform complements this by analyzing and visualizing data generated by the OrthoAI system.

6.1.1 OrthoAI System

OrthoAI is an AI-driven platform designed to answer orthopedic-related questions. It provides users with insightful, accurate answers by leveraging machine learning and natural language processing technologies to process the questions and generate responses.

Key Features of OrthoAI:

- **AI-Powered Question Answering:** The core function of OrthoAI is to answer questions related to orthopedic issues. These could range from joint pain, fractures, injuries, and other bone-related topics.
- **Real-Time Response Generation:** OrthoAI generates real-time answers by analyzing user-submitted questions and referencing a knowledge base of orthopedic medical information.
- **Medical Data Integration:** The system is continuously updated with the latest research in orthopedics to ensure that answers are relevant and up-to-date.

OrthoAI does not involve complex categorization or synonym detection directly but instead focuses on providing immediate, AI-generated responses to user queries related to orthopedic topics.

6.1.2 Analytics Platform

The Analytics platform is an essential system for collecting and analyzing the data generated by the OrthoAI system. It tracks system usage, feedback, and identifies trends in orthopedic concerns based on user queries. The Analytics platform also allows us to analyze common issues, improve the AI system, and monitor its performance over time.

Key Features of the Analytics Platform:

- **Query Categorization:** The platform categorizes common questions into domains like *muscle-related issues*, *bone structure*, *neural disorders*, and *injuries*.
- **Synonym Detection and Question Standardization:** By analyzing the frequency of various question formulations, the platform standardizes queries, helping the system better understand diverse ways of asking the same question.
- **User Feedback Analysis:** User ratings and feedback are tracked and analyzed to improve the performance and accuracy of the system. The platform helps to identify which answers require refinement or improvement.
- **Data Visualization and Reporting:** Analytics helps visualize trends in data, such as which types of orthopedic issues are being queried the most, allowing for better decision-making and system optimization.

The Analytics platform works hand-in-hand with OrthoAI, providing insights into usage patterns, system performance, and user preferences.

6.2 Tools and Technologies Used

The development of OrthoAI and its Analytics platform involved several technologies and tools that ensured robustness, scalability, and performance across the entire system.

6.2.1 OrthoAI System Backend

- **Node.js:** Used to build a scalable and efficient backend API, responsible for processing incoming queries and providing answers from the AI model.
- **Prisma ORM:** Used for database interaction, allowing for easy querying and database management to store user questions and answers.
- **PostgreSQL:** Chosen for storing structured data like user queries, answers, and system metadata. PostgreSQL ensures reliability and scalability for high-traffic systems.
- **Redis:** Used for caching frequently accessed data to improve system performance and reduce query response time.

6.2.2 Machine Learning for OrthoAI

- **Python:** Python was used to develop machine learning models that process orthopedic-related questions.
- **spaCy:** A powerful NLP library used for text preprocessing and understanding user queries in natural language.
- **Transformers (Hugging Face):** Transformer-based models are employed for generating real-time answers to orthopedic-related questions.

6.2.3 Analytics Platform Tools

- **Power BI/Tableau:** Used for generating interactive dashboards and data visualizations that display query trends, user activity, and system performance metrics.
- **Elasticsearch:** Utilized for fast indexing and searching of large datasets, enabling efficient querying and analytics.
- **Grafana:** Used to monitor the health and performance of the OrthoAI platform in real-time, ensuring optimal uptime and reliability.

6.3 System Design and Architecture

The architecture of the OrthoAI system is designed for scalability, modularity, and resilience. The system handles AI-driven question answering, while the Analytics platform monitors, categorizes, and visualizes data collected by OrthoAI.

6.3.1 High-Level Architecture for OrthoAI

- **Frontend Layer:** Built with React and Next.js, the frontend enables users to submit queries and view responses generated by OrthoAI.
- **Backend Layer:** The backend, powered by Node.js, handles user queries, processes them with machine learning models, and returns AI-generated answers.
- **Machine Learning Layer:** The AI models, trained on orthopedic medical data, process user queries and generate responses in real-time.

- **Database Layer:** PostgreSQL stores structured data, such as query logs and user feedback, while Redis caches frequently accessed information for faster response times.

6.3.2 Analytics Platform Architecture

- **Data Collection Layer:** This layer collects query data, user feedback, and system metrics for processing.
- **Processing and Storage Layer:** Data is stored in a structured format, with Elasticsearch helping to index and search the data efficiently.
- **Visualization Layer:** The data is visualized in real-time on dashboards powered by Power BI, Tableau, and Grafana.

6.3.3 Data Flow for OrthoAI

- A user submits a query through the frontend.
- The backend processes the query and generates a real-time answer using the AI model.
- The answer is returned to the frontend and displayed to the user.

6.3.4 Data Flow for Analytics

- Query data and user feedback are collected from the frontend.
- The data is processed and stored for analytics.
- Analytics tools like Power BI and Grafana visualize trends and metrics, providing actionable insights into system performance and user behavior.

6.4 Challenges Faced and Solutions

6.4.1 Challenge 1: Managing Multiple Query Formulations

Different users may ask the same question in different ways. Standardizing these queries was essential for OrthoAI's success. By integrating NLP models that can detect and normalize question formulations, we ensured better query processing.

6.4.2 Challenge 2: Real-Time Answer Generation

Generating accurate, real-time answers for unseen queries was challenging. Transformer models, pre-trained on orthopedic-specific datasets, helped solve this issue by generating contextually accurate answers.

6.4.3 Challenge 3: Scalability and Performance

Handling high traffic and ensuring fast response times required careful system architecture planning. The use of Redis for caching and optimized queries in PostgreSQL ensured smooth scaling and reduced latency during high traffic periods.

6.5 Future Work

Future work will include:

- **Enhanced AI Models:** Further improvements to the AI models for more accurate real-time answer generation.
- **Multilingual Support:** Expanding the platform's capabilities to support multiple languages for a global audience.
- **Continuous Research Integration:** Ongoing updates to the medical knowledge base, ensuring that OrthoAI stays up-to-date with the latest orthopedic research.

CHAPTER 7

SOFTWARE TESTING

7.1 Types of Testing

Comprehensive testing was conducted throughout the development lifecycle to ensure the reliability, accuracy, and usability of the Ortho AI system. Various testing methodologies were applied at different stages, covering both the frontend, backend, and model components.

7.1.1 Unit Testing

Each software module was independently tested to verify its functionality:

- **Login Module:** Tested for secure authentication, input validation, and error messages on incorrect credentials.
- **User Query Module:** Verified proper capturing of user inputs, language normalization, and query submission flow.
- **AI Response Module:** Ensured accurate AI-generated answers, proper mapping with medical tags, and fallback mechanisms.
- **Feedback Module:** Tested logging of user feedback, correctness flags, and dashboard reflection.
- **Dashboard Analytics Module:** Validated data aggregation, location-wise trends, and disease categorization display.
- **Data Storage Module:** Verified correct storage of queries, responses, and metadata using Prisma with PostgreSQL.

7.1.2 Integration Testing

Integration tests were carried out to ensure seamless interaction between modules:

- Verified that frontend API calls correctly interacted with the backend services.
- Ensured the data flow from user query submission to AI answer retrieval and logging was consistent.
- Cross-checked feedback mechanism linking backend storage and frontend display.

7.1.3 Functional Testing

All major features were tested against functional requirements:

- AI response accuracy and delivery.
- User authentication and session handling.
- Real-time feedback submission by doctors.
- Admin dashboard operations for monitoring model performance.

7.1.4 Performance Testing

Stress testing was conducted to evaluate system performance:

- Simulated a high number of concurrent queries to assess backend and model scalability.
- Tested database efficiency with bulk inserts and queries.
- Evaluated frontend load time and responsiveness under heavy load.

7.1.5 User Acceptance Testing (UAT)

End-user scenarios were simulated to validate system usability and functionality:

- Doctors tested AI responses and flagged any inaccuracies.
- Admins tested dashboard functionalities including filtering, sorting, and exporting reports.
- General usability and intuitiveness of the UI were assessed.

7.2 Test Cases & Test Results

Test Case ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-001	API Backend	Submit a user query via frontend	Query received and processed, AI response generated	Response successfully generated and returned	Pass
TC-002	Frontend UI	Login and navigate between dashboard pages	Smooth navigation without crashes or unexpected behavior	Navigation successful	Pass
TC-003	Database	Store and retrieve AI responses with metadata	Accurate insertion and retrieval of query-response pairs	Correct data stored and fetched	Pass
TC-005	Model Output Validation	Doctors manually review AI-generated answers	Correct, contextually relevant responses	Responses validated by experts	Pass
TC-006	Feedback Mechanism	Submit feedback on AI responses	Feedback recorded and visible in the admin dashboard	Feedback submission successful	Pass
TC-007	Authentication	Test login, logout, and session expiration	Secure login, proper logout, session timeout after inactivity	Authentication works as expected	Pass
TC-008	Load Handling	Simulate 500+ concurrent user queries	System handles load without crashing or major delays	Handled successfully	Pass
TC-009	Dashboard Reports	Export AI response statistics	Downloadable CSV report with correct data	Report generated correctly	Pass

Table 7.1: Test Case Results for Ortho AI System

CHAPTER 8

RESULTS

8.1 Outcomes

1. Outcome 1
1. Outcome 2
1. Outcome 3

8.2 Screen Shots

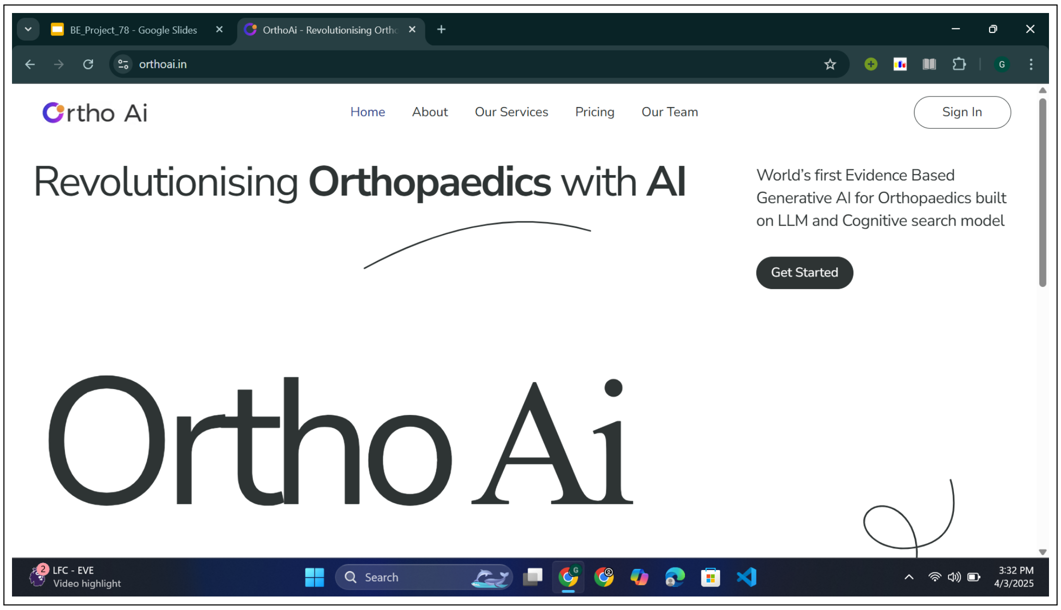

Figure 8.1: Home Page

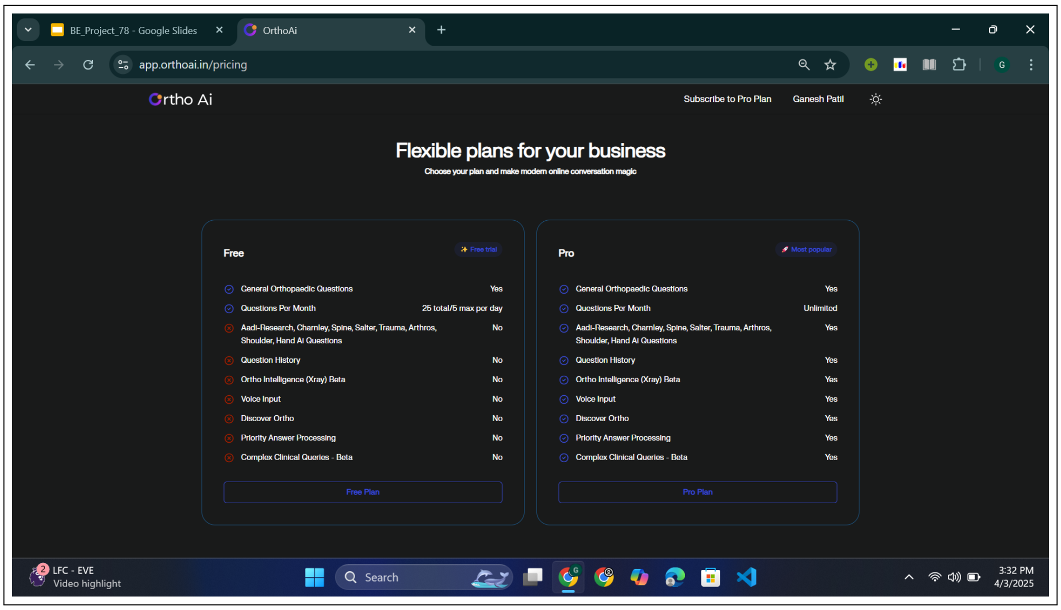

Figure 8.2: Pricing

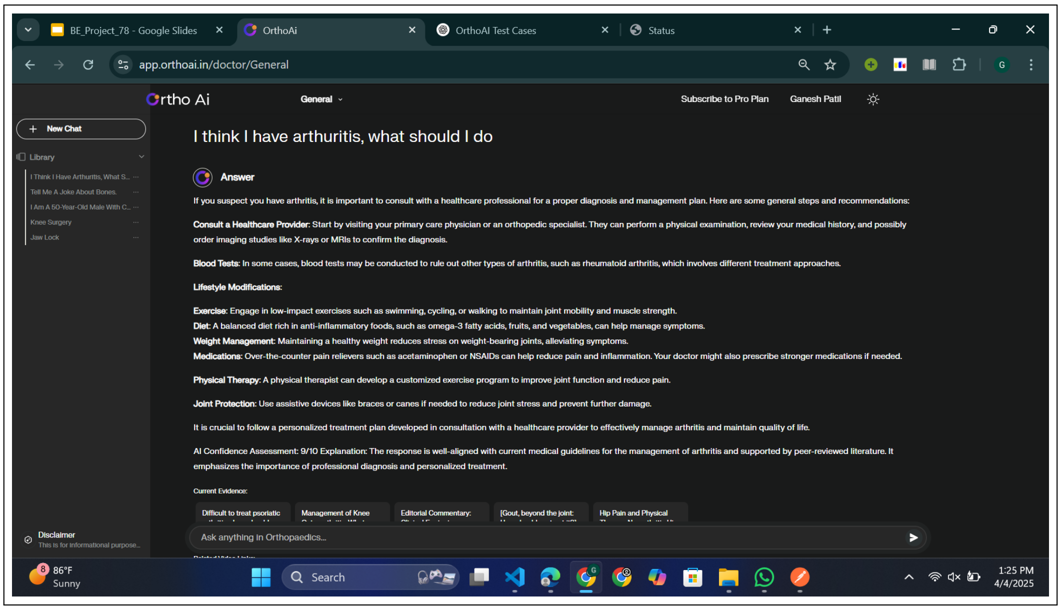

Figure 8.3: Query results

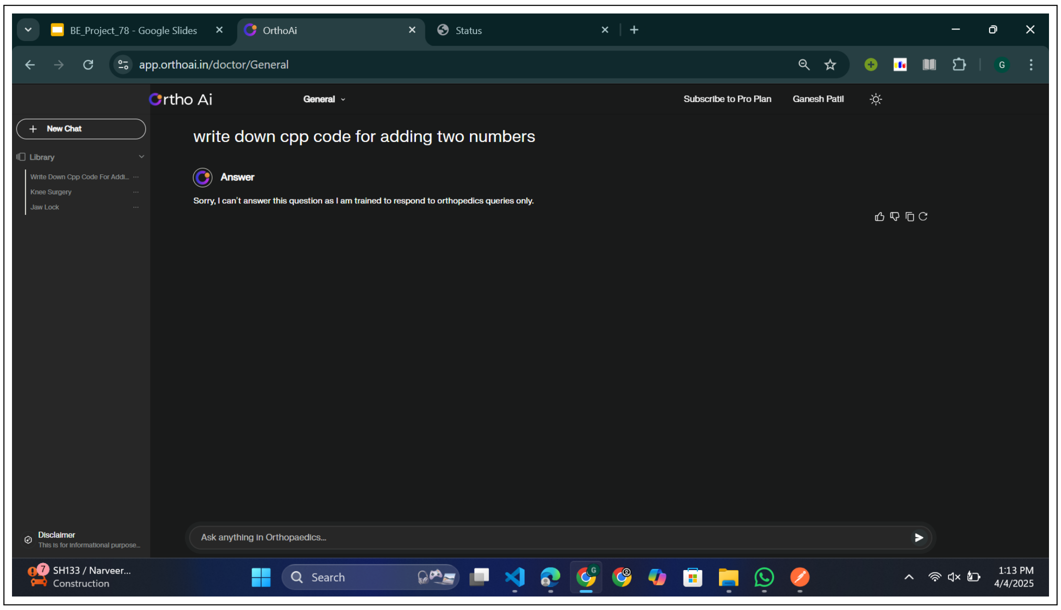

Figure 8.4: Simple Jailbreaking

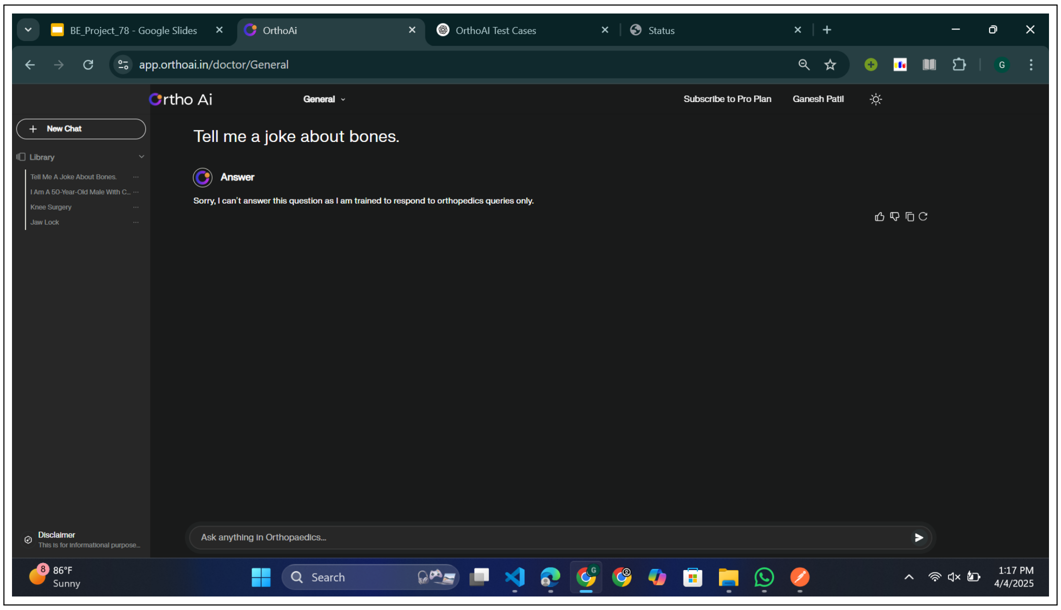

Figure 8.5: Complex Jailbreaking

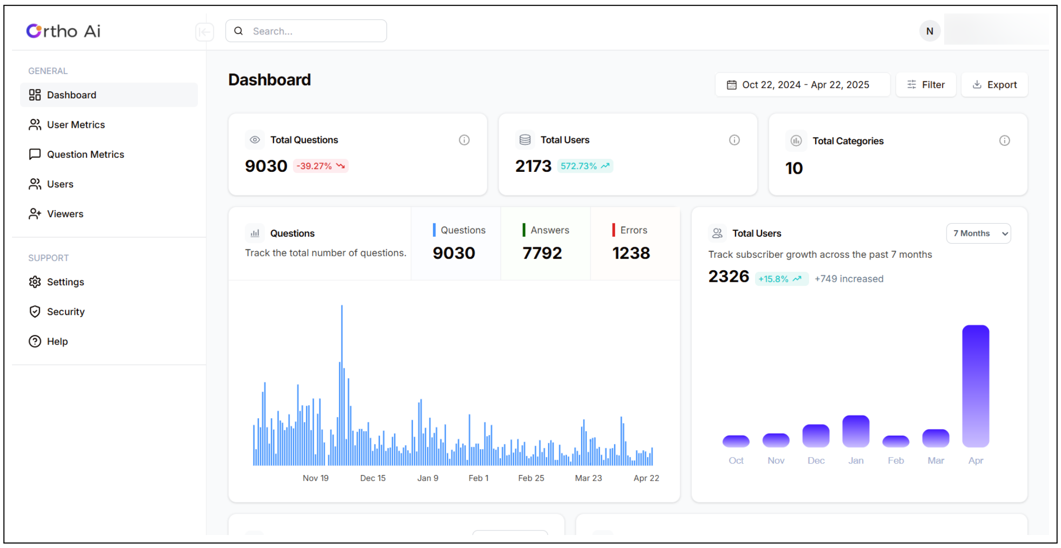

Figure 8.6: Analytics Dashobard

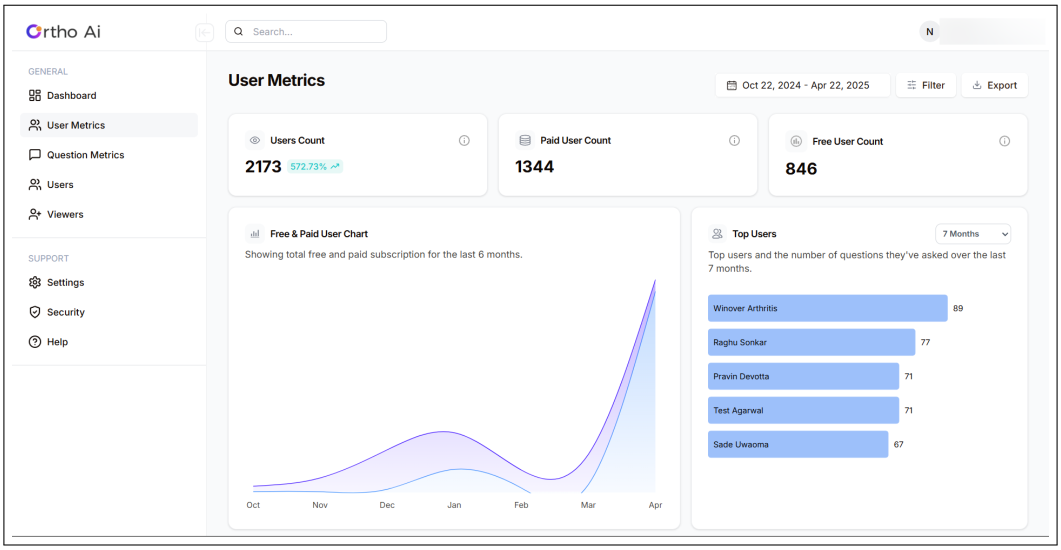

Figure 8.7: User Analysis

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

The project successfully delivers a dual-purpose intelligent system combining a domain-specific AI answering engine—**OrthoAI**—with an automated file categorization and anomaly detection framework.

OrthoAI is designed to answer orthopedic-related questions using transformer-based language models. It standardizes various question formats, categorizes them into medical domains like muscle, bone, neural, and injury-related queries, and provides dynamic, contextual answers. By leveraging real-time language model inference and continuous research integration, OrthoAI ensures its medical responses remain accurate and up-to-date. It also implements a feedback mechanism that allows users to rate the responses, enabling iterative model improvement.

Parallelly, the intelligent file monitoring system accurately classifies files based on extensions and content using machine learning techniques while also tracking abnormal file behavior. This includes real-time anomaly detection and explainable alerts using XAI (Explainable AI).

Together, both systems enhance organizational intelligence—OrthoAI improves accessibility to orthopedic knowledge, and the file system ensures security and governance in data management environments.

9.2 Future Work

9.2.1 Personalized Query Understanding in OrthoAI

To improve OrthoAI's adaptability, future enhancements will include user persona-based query processing. Questions asked by doctors, students, or patients will be interpreted and answered differently to suit their expertise levels and context.

9.2.2 Domain Expansion

Currently focused on orthopedics, OrthoAI can be extended to other medical domains such as cardiology, neurology, or physiotherapy, turning it into a general-purpose medical Q&A assistant.

9.2.3 Integration into Healthcare Platforms

OrthoAI will be packaged as an API for seamless integration into:

- Telemedicine platforms

- Hospital management systems
- Electronic Medical Record (EMR) systems

9.2.4 Real-Time Remediation and Response Mechanisms (File Monitoring)

For file anomaly detection, future systems will support:

- Blocking unauthorized file access
- Quarantining suspicious files
- Triggering access control policies

This will transition the system into an active defense mechanism.

9.2.5 Continuous and Online Learning Models

Both modules will implement adaptive models:

- OrthoAI will learn from user ratings and feedback in real-time.
- The file system will evolve with online anomaly detection and file type recognition.

9.2.6 Multi-Platform and Cloud Support

The file monitoring module will support hybrid environments, including:

- Google Drive
- OneDrive
- Dropbox
- AWS S3

Meanwhile, OrthoAI will also be deployed as a cloud service, allowing scalable access for healthcare institutions.

9.3 Applications

- **Orthopedic Virtual Assistant:** Real-time AI-driven Q&A for orthopedic queries.
- **Clinical Decision Support:** Categorized and research-backed insights for doctors.
- **Dashboard Analytics:** Location-based trends and disease-specific query tracking.
- **Hospital Integration:** Assistive tool for clinics and hospitals to handle patient FAQs.
- **Medical Education Aid:** Quick reference for students and trainees in orthopedics.
- **Healthcare Marketing Insights:** Data-driven targeting based on regional health trends.

CHAPTER 10

APPENDIX

10.1 Appendix A

Problem Statement Feasibility Assessment Using Satisfiability Analysis, Complexity Theory, and Modern Algebra

This appendix evaluates the feasibility of the project titled **"AI-Driven Healthcare Solution – OrthoAI"** by analyzing logical satisfiability, computational complexity, and modern algebraic models relevant to the system's architecture and functionality.

1. Satisfiability Analysis

Phase 1 – Evidence-Based Query Resolution:
The system uses a generative AI model grounded in a curated vector database comprising peer-reviewed orthopaedic literature (e.g., PubMed, YouTube). For well-defined queries related to common orthopaedic conditions, the AI consistently provides satisfiable, accurate responses. However, the satisfiability can degrade with ambiguous or poorly structured queries, requiring robust natural language understanding and contextual inference.

Phase 2 – Surgical Decision Support:
The system offers real-time intraoperative support, including stepwise procedural guidance, identification of possible complications, and instrument suggestions. While standard procedures follow well-defined workflows and are satisfiable through learned surgical patterns, rare or complex surgical cases may exhibit non-determinism, introducing computational challenges due to intraoperative variations and patient-specific anomalies.

2. Application of Modern Algebra

Group Theory:
Group theory is used to model state transitions in surgical workflows and patient treatment paths. Each surgical step or decision can be represented as a transformation that maintains certain invariants (e.g., patient safety, procedural integrity). This formalism helps maintain consistency across multiple surgical paths and supports reversibility in the decision process where applicable.

3. Computational Complexity Classification

Task	Complexity Class	Justification
Evidence-Based Query Resolution	**P (Polynomial)**	Retrieval is performed using efficient vector similarity search; LLM inference is deterministic once input is vectorized.
Handling Ambiguous/Edge Queries	**NP-Complete**	Requires contextual inference across overlapping medical concepts; LLM behavior may vary based on ambiguous inputs.
Real-Time Surgical Decision Support	**NP-Hard**	Involves real-time decision-making, uncertainty, and dynamic changes that are computationally complex to model fully.

Table 10.1.3: Complexity Classification of OrthoAI Project Tasks

4. Conclusion

The proposed AI-Driven Healthcare Solution, OrthoAI, is practically feasible within well-defined operational constraints:

- Evidence-based query resolution is computationally efficient and reliable through LLM integration with a domain-specific vector database.
- Surgical decision support introduces higher computational complexity due to real-time dynamics and the need for adaptive reasoning in uncertain environments.

The system is intentionally constrained to provide orthopaedic-related information, ensuring focus and accuracy in this specialized domain. The use of algebraic modeling contributes structure and consistency to procedural decision-making. This feasibility study supports the viability of the system, provided that optimizations, clinical heuristics, and scalable architectures are incorporated for real-time performance and accuracy.

10.2 Appendix B: Details of Paper Publication

Paper Title: *AI-Driven Healthcare Solution – OrthoAI: Evidence-Based Query Resolution and Surgical Decision Support*

1. Name of the Conference/Journal
Conference/Journal Name: *Tijer's International Journal of Emerging Research in Technology & Innovation*
Publication Link: `https://tijer.org/tijer/track.php?r_id=156229`

2. Reviewers' Comments
Reviewer 1: "The approach presented in this paper provides a novel and impactful solution in the healthcare domain, particularly in orthopaedics. The integration of a domain-specific vector database with semantic similarity search is well-suited for evidence-based query resolution."
Reviewer 2: "The real-time surgical decision support feature is promising but could benefit from more extensive validation in actual clinical environments. Future work should focus on scalability and robustness in dynamic and varied surgical settings."
Reviewer 3: "A well-structured paper with solid technical background and clarity. I would recommend additional details on the system architecture and real-world testing to strengthen the claims of real-time efficacy."

3. Paper
Citation:
Vishal Ananda Kuwar, Parth Mahesh Asawa, Ganesh Ravindra Patil, Vedant Nandkumar Pawar, Preeti Jain, Amit Yedurkar, "AI-Driven Healthcare Solution – OrthoAI: Evidence-Based Query Resolution and Surgical Decision Support," *Tijer's International Journal of Emerging Research in Technology & Innovation*, 2025.

4. Certificate
Certificate of Publication: `https://tijer.org/tijer/certificatemanager.php?a_rid=156229`

10.3 Appendix C

Plagiarism Report of project report.

Document Information

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ABSTRACT This project AI Driven Healthcare Solution, an intelligent biomedi- cal question-answering system tailored for orthopaedic-related inquiries. Lever- aging domain-specific literature from PubMed, the system employs NeuML's PubMedBERT embeddings to convert textual data into high-dimensional vec- tor representations. These embeddings are stored in Qdrant, a vector database optimized for similarity search. Upon receiving a user query, OrthoAI gen- erates a corresponding embedding and retrieves the top seven semantically similar documents. These documents are then processed using a generative language model to synthesize coherent and contextually relevant responses. Complementing its core functionality, OrthoAI integrates a comprehen- sive analytics dashboard that facilitates real-time monitoring of user inter- actions and system performance. The dashboard tracks metrics such as geo- graphic distribution of queries, frequency of procedure-specific searches, sys- tem errors, token usage anomalies, user activity trends, and subscription- based user segmentation. This integration of natural language processing with operational analytics underscores the potential of domain-specific AI systems in enhancing accessibility to biomedical knowledge. CHAPTER 1 INTRODUCTION

AI Driven Healthcare Solution 1.1 Overview Modern healthcare increasingly relies on the integration of intelligent systems to streamline information access and deliver precise insights. In the ortho- pedic domain, where practitioners frequently require up-to-date, research- backed information, the ability to retrieve relevant content rapidly is essen- tial. Traditional search engines and static databases are often inefficient for real-time, context-aware query resolution. This project introduces an AI- driven orthopedic assistant designed to respond accurately to medical queries by leveraging biomedical literature as its core data source. The assistant uti- lizes pre-trained biomedical embeddings on a curated dataset of PubMed research papers to extract meaningful semantic patterns and provide high- precision responses. Beyond query resolution, the system incorporates a comprehensive analytics dashboard that presents usage trends, system per- formance metrics, and error tracking, enabling effective decision-making and user behavior monitoring. By organizing and visualizing this health-related query data, the dashboard acts as a powerful business intelligence tool, en- abling decision-makers to: Detect rising health concerns in specific regions, Identify trends in symptom frequency over time, Pinpoint demand for partic- ular treatments or medication categories, Uncover seasonal or location-based health issues 1.2 Motivation Orthopedic professionals frequently consult medical literature to validate clinical decisions. However, manually searching through dense academic con- tent is time-consuming and error-prone. General-purpose AI tools often lack the domain specificity required for orthopedic cases, while existing clinical databases are not optimized for conversational interactions or contextual search. Additionally, there is a growing need to monitor the operational aspects of AI services—such as usage volume, geographic distribution, and error diagnostics—to support scalability and ensure quality of service. The motivation behind this project is to address these two dimensions: enhanc- ing medical query responses with domain-tuned AI systems and offering a real-time analytics interface for evaluating system usage and impact. PICT, Department of Computer Engineering 2024-25 Page 2

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OrthoAI: A Domain-Specific Biomedical QA System with Real-Time Analytics 1 st Vishal Ananda Kuwar

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CHAPTER 11

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LIST OF ABBREVATIONS

Abbreviation	Illustration
AI	Artificial Intelligence
LLM	Large Language Model
BI	Business Intelligence
API	Application Programming Interface

LIST OF FIGURES

LIST OF TABLES